

Preprocessing Pipeline Documentation

Project Title

Enterprise Data Preparation for Employee Attrition

Purpose

The preprocessing pipeline was created to automate the data preparation process. Instead of manually cleaning and transforming the dataset every time, the pipeline applies the same preprocessing steps consistently.

This approach is useful in enterprise environments because it makes the data preparation process more reliable, repeatable, and easier to use with future datasets.

Pipeline Components

The preprocessing pipeline contains two main sections:

Numerical Data Pipeline

The numerical pipeline performs the following steps:

1. Missing values are replaced using the median strategy.
2. Numerical variables are standardized using StandardScaler.

The median imputation step ensures that the pipeline can also handle future datasets containing missing numerical values.

Categorical Data Pipeline

The categorical pipeline performs the following steps:

1. Missing categorical values are replaced using the most frequent value.
2. Categorical variables are converted into numerical features using One-Hot Encoding.
3. Unknown categories are safely handled without causing an error.

This makes the pipeline suitable for processing new employee data.

Column Transformation

ColumnTransformer is used to apply different preprocessing operations to numerical and categorical columns.

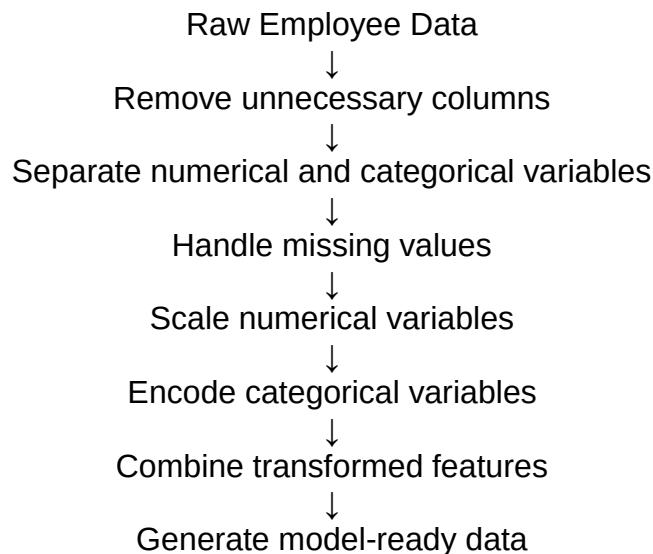
Numerical columns are sent to the numerical pipeline, while categorical columns are sent to the categorical pipeline. Both transformed outputs are then combined into a single processed dataset.

Benefits of the Pipeline

The preprocessing pipeline provides several benefits:

- Ensures consistent data preparation.
- Reduces manual work.
- Handles missing values automatically.
- Scales numerical features.
- Encodes categorical features.
- Supports future employee datasets.
- Reduces the possibility of data processing errors.
- Makes the project easier to integrate with machine learning models.

Pipeline Workflow



Conclusion

The preprocessing pipeline successfully automates the major data preparation tasks required for the Employee Attrition project. It provides a structured and reusable method for transforming raw HR data into machine-learning-ready data.

The pipeline can also be extended in the future by connecting it directly with classification models such as Logistic Regression, Random Forest, or other machine learning algorithms.